Spring 2015

Marmara University
Electrical and Electronics Engineering
STAT253-Probability and Statistics
Final Exam

1. Let the discrete random variables X and Y have the following joint probability mass function (PMF)

$$p_{X,Y}[i,j] = \frac{i+j}{21} \qquad i = 1,2,3, \quad j = 1,2$$

(a) Calculate the marginal PMFs $p_X[i]$ and $p_Y[j]$. [5 pts.]
(b) Calculate $p_{Y|X}[j|X = 3]$, the conditional PMF of Y given $X = 3$. [10 pts.]
(c) Calculate $E[Y|X = 3]$ and $\mathrm{VAR}(Y|X = 3)$. [10 pts.]

2. Let the continuous random variables X and Y have the following joint probability density function (PDF)

$$p_{X,Y}(x,y) = \begin{cases} 2, & 0 \le x \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

(a) Compute $P(\frac{3}{4} \le Y \le \frac{7}{8} | X = \frac{1}{4})$. [10 pts.]
(b) Find the correlation coefficient of X and Y. [15 pts.]

3. Let the discrete random variable X have the following moment generating function

$$\Phi_X[\omega] = [0.25 + 0.75e^{\omega}]^{12}$$

(a) Compute $E[X]$. [10 pts.]
(b) What is $\mathrm{VAR}(X)$. [15 pts.]

4. A continuous random variable X has the following PDF

$$p_X(x) = \begin{cases} 2(1-x), & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find $P(\frac{1}{4} < X < \frac{3}{4})$. [10 pts.]
(b) Calculate $E[X]$ and $\mathrm{VAR}(X)$. [15 pts.]

Directions:

- One A4-sized formula sheet (without solutions) can be used.
- The exam is 75 minutes.

GOOD LUCK!

Spring 2025

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1.

$y \backslash x$	1	2	3
1	$\frac{2}{21}$	$\frac{3}{21}$	$\frac{4}{21}$
2	$\frac{3}{21}$	$\frac{4}{21}$	$\frac{5}{21}$

(5 pts)

(a)

x	1	2	3
$P_X(x)$	$\frac{5}{21}$	$\frac{7}{21}$	$\frac{9}{21}$

y	1	2
$P_Y(y)$	$\frac{9}{21}$	$\frac{12}{21}$

(10 pts)

(b) $P_{Y|X}[y|X=3] = \frac{P_{X,Y}(y, X=3)}{P_X(X=3)} = \begin{cases} \frac{4/21}{9/21}, & y=1 \\ \frac{5/21}{9/21}, & y=2 \end{cases} = \begin{cases} \frac{4}{9}, & y=1 \\ \frac{5}{9}, & y=2 \end{cases}$

(10 pts)

(c) $E[Y|X=3] = \sum_{y=1}^{2} y\, P_{Y|X}(y|X=3) = 1\cdot\frac{4}{9} + 2\cdot\frac{5}{9} = \frac{14}{9}$

$E[Y^2|X=3] = \sum_{y=1}^{2} y^2 \cdot P_{Y|X}(y|X=3) = 1^2\cdot\frac{4}{9} + 2^2\cdot\frac{5}{9} = \frac{24}{9}$

(10 pts)

$\text{var}(Y|X=3) = \frac{24}{9} - \left(\frac{14}{9}\right)^2 = \frac{24}{9} - \frac{196}{81}$

2. (a) $P_X(x) = \int_x^1 P_{X,Y}(x,y)\,dy = \int_x^1 2\,dy = 2y\Big|_x^1 = 2(1-x),\ 0\le x\le 1$

$P_{Y|X}(y|x) = \frac{P_{X,Y}(x,y)}{P_X(x)} = \frac{2}{2(1-x)} = \frac{1}{1-x},\ 0\le y\le 1$

(10 pts)

$P_{Y|X}(y|X=\frac{1}{4}) = \frac{1}{1-\frac{3}{4}} = \frac{4}{3},\ \frac{1}{4}\le y\le 1$

$P\left(\frac{3}{4}\le Y\le\frac{7}{8}\,\middle|\,X=\frac{1}{4}\right) = \int_{3/4}^{7/8}\frac{4}{3}\,dy = \frac{4}{3}\left(\frac{7}{8}-\frac{3}{4}\right) = \frac{4}{3}\cdot\frac{1}{8} = \frac{1}{6}$

(b) $P_X(x) = 2(1-x),\ 0\le x\le 1$ from part (a)

(3 pts) $P_Y(y) = \int_0^y P_{X,Y}(x,y)\,dx = \int_0^y 2\,dx = 2y,\ 0\le y\le 1$

(3 pts) $E[XY] = \int_0^1\int_0^y xy\,2\,dx\,dy = \int_0^1 2y\left[\int_0^y x\,dx\right]dy = \int_0^1 2y\left[\frac{x^2}{2}\Big|_0^y\right]dy = \int_0^1 y^3\,dy = \frac{y^4}{4}\Big|_0^1 = \frac{1}{4}$

$E[X] = \int_0^1 x\cdot 2(1-x)\,dx = \int_0^1 (2x-2x^2)\,dx = \frac{2x^2}{2} - \frac{2x^3}{3}\Big|_0^1 = 1-\frac{2}{3} = \frac{1}{3}$

$E[X^2] = \int_0^1 x^2\cdot 2(1-x)\,dx = \int_0^1 (2x^2-2x^3)\,dx = \frac{2x^3}{3} - \frac{2x^4}{4}\Big|_0^1 = \frac{2}{3}-\frac{2}{4} = \frac{1}{6}$

(3 pts) $\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{1}{6} - \left(\frac{1}{3}\right)^2 = \frac{1}{6} - \frac{1}{9} = \frac{1}{18}$

$E[Y] = \int_0^1 y\cdot 2y\,dy = \frac{2y^3}{3}\Big|_0^1 = \frac{2}{3}$

$E[Y^2] = \int_0^1 y^2\cdot 2y\,dy = \frac{2y^4}{4}\Big|_0^1 = \frac{2}{4}$

(3 pts) $\text{var}(Y) = E[Y^2] - (E[Y])^2 = \frac{2}{4} - \left(\frac{2}{3}\right)^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$

$$\rho_{X,Y} = \frac{E[XY] - E[X]\cdot E[Y]}{\sqrt{\text{var}(X)\cdot \text{var}(Y)}} = \frac{\frac{1}{4} - \frac{1}{3}\cdot\frac{2}{3}}{\sqrt{\frac{1}{18}\cdot\frac{1}{18}}} = \frac{\frac{1}{36}}{\frac{1}{18}} = \frac{18}{36} = \frac{1}{2}$$ (8 pts)

3. (a) $\frac{d\phi_X(\omega)}{d\omega} = 12[0.25 + 0.75e^{j\omega}]^{11}\cdot j0.75e^{j\omega} = j\times 9\cdot[0.25+0.75e^{j\omega}]^{11}\cdot e^{j\omega}$ (5 pts)

(5 pts) $\frac{d^2\phi_X(\omega)}{d\omega^2} = j\times 9\times 11[0.25+0.75e^{j\omega}]^{10}\cdot j0.75e^{j\omega}\cdot e^{j\omega} + j\cdot j\cdot 9e^{j\omega}[0.25+0.75e^{j\omega}]^{11}$

$= j^2\cdot 9\cdot 11\cdot 0.75[0.25+0.75e^{j\omega}]^{10}\cdot e^{j2\omega} + j^2\cdot 9[0.25+0.75e^{j\omega}]^{11}\cdot e^{j\omega}$

$E[X] = \frac{1}{j}\left.\frac{d\phi_X(\omega)}{d\omega}\right|_{\omega=0} = \left.9[0.25+0.75e^{j\omega}]^{11}\cdot e^{j\omega}\right|_{\omega=0} = 9[0.25+0.75]^{11} = 9$ (5 pts)

(b)
$E[X^2] = \frac{1}{j^2}\left.\frac{d^2\phi_X(\omega)}{d\omega^2}\right|_{\omega=0} = 9\times 11\times 0.75 + 9 = \frac{333}{4}$ (5 pts)

$\text{VAR}(X) = E[X^2] - (E[X])^2 = \frac{333}{4} - 81 = \frac{9}{4}$ (5 pts)

4. (a) $P\left(\frac{1}{4} < X < \frac{3}{4}\right) = \int_{1/4}^{3/4} P_X(x)\,dx = \int_{1/4}^{3/4} 2(1-x)\,dx = 2x - x^2\Big|_{1/4}^{3/4} = \frac{8}{16} = \frac{1}{2}$ (10 pts)

(b) $E[X] = \int_0^1 x\,2(1-x)\,dx = \int_0^1 (2x - 2x^2)\,dx = \frac{1}{3}$ (5 pts)

$E[X^2] = \int_0^1 x^2\,2(1-x)\,dx = \int_0^1 (2x^2 - 2x^3)\,dx = \frac{1}{6}$ (5 pts)

$\text{VAR}(X) = E[X^2] - (E[X])^2 = \frac{1}{6} - \left(\frac{1}{3}\right)^2 = \frac{1}{18}$ (5 pts)